	GLOBAL ECONOMICS **Grade: 11th - Learning Evidence 3** **Statistical models and research planning** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Apply continuous probability models and sampling distributions to investigate economic and financial questions, justify statistical methods, and communicate evidence-based conclusions.	**Assessment criteria:** C8: Uses Student's t-distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges. C9: Calculates and compares critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample. C10: Integrates continuous probability modelling, simulation, sampling analysis, and portfolio-risk measures to investigate an economic or financial question, justifying the selected methods and communicating results, assumptions, limitations, and evidence-based conclusions.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

Show all relevant calculations and clearly justify your answers. Partial credit may be awarded when the correct procedure is clearly demonstrated, even if arithmetic errors are made. No points will be awarded for answers without a demonstrated procedure, except for the brief research selection in Question 7.

A minimum of 4 points is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

Question	C8	C9	C10
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7	–	–	6
Total	**6**	**6**	**6**

Answer on separate paper and show your reasoning. Use the supplied normal and Student's t reference sheets. Report Student's t critical values and normal critical values from the infinity row to **3 decimal places**, and normal probabilities to **4 decimal places**. For Student's t cutoffs between printed entries, give probability bounds without interpolation. Each criterion is worth **6 points** (**18 points** total); at least **4 points per criterion** are required to pass that criterion.

Questions 1–6 concern independent random observations from normal populations. All critical values and probability questions refer to the sample mean standardized relative to the population mean; critical values have no units.

1. Export inspection costs [C8-2]

An exporter studies average inspection costs using 12 shipments. The population standard deviation is unknown and estimated from the sample.

(a) Determine the degrees of freedom and the positive critical value that leaves 1 percent of the probability to its right. **[1]**

(b) Determine the negative critical value below which 5 percent of the probability lies. Justify its sign and the table column used. **[1]**

2. Market-stall electricity use [C8-2]

A study investigates average electricity use across 25 market stalls. The population standard deviation is unknown and estimated from the sample.

(a) Determine the degrees of freedom. For a central region containing 98 percent of the probability, determine the combined probability outside it and the probability in each tail. **[1]**

(b) Determine the 2 critical values, symmetric about 0, enclosing this central region. Identify the table column used. **[1]**

3. Cold-storage energy costs [C8-2]

An analyst studies average energy costs using 15 cold-storage facilities. The population standard deviation is unknown and estimated from the sample.

(a) Determine the degrees of freedom and give bounds for the probability that the standardized sample mean exceeds positive 2.40. **[1]**

(b) Give bounds for the probability that the standardized sample mean is below negative 2.40, and for the probability that it lies between negative 2.40 and positive 2.40. **[1]**

4. Bottling-line measurements [C9-2]

A bottling plant studies average bottle volume using 16 bottles. The population standard deviation is known to be 3 millilitres.

(a) Choose and justify the distribution model for the standardized sample mean. Determine the negative critical value below which 2.5 percent of the probability lies. **[1]**

(b) Calculate the probability that the standardized sample mean lies between negative 1.00 and positive 1.50. **[1]**

5. Repair-time averages [C9-2]

A workshop studies average repair time using 10 repairs. The population standard deviation is unknown; the sample standard deviation is 4 minutes.

(a) Choose and justify the distribution model, determine the degrees of freedom, and find the positive critical value that the standardized sample mean exceeds with probability 2.5 percent. **[1]**

(b) Estimate the probability that the standardized sample mean lies between negative 1.833 and positive 2.821. **[1]**

6. Comparing parcel-mass studies [C9-2]

2 studies investigate average parcel mass. Study A uses 12 parcels and a known population standard deviation of 0.4 kilograms. Study B uses 12 parcels; its population standard deviation is unknown and estimated by a sample standard deviation of 0.4 kilograms.

(a) Justify the distribution model for each study's standardized sample mean. Determine each model's 2 critical values, symmetric about 0, enclosing the central 95 percent of the probability. Calculate the difference between the positive critical values and explain it in terms of tail behaviour. **[1]**

(b) Study B increases its sample to 61 parcels, with population standard deviation still unknown. State its model and degrees of freedom, and determine its new critical values enclosing the central 95 percent of the probability. Explain the trend toward the normal model and why sample size alone does not determine the model choice. **[1]**

7. Choose your research option [C10-6]

During the exam: select **1** of the following options and write a very brief description (1–2 sentences) of real-life data that could plausibly follow your selected distribution. **[6]**

(a) A continuous variable modelled by a linear probability density over a stated range.

(b) Waiting times or intervals between events modelled by an exponential distribution.

(c) Normally distributed measurements.

No datasets, sources, calculations, parameter estimates, or detailed model justification are required during the exam.

Online report submission: your PDF report will focus on the **1** selected option and must be submitted virtually through **Open**. This online submission is separate from the brief response written during the exam. To demonstrate the full C10 criterion, investigate an economic or financial question using real data and your selected continuous model. Include simulation, sampling analysis, and portfolio-risk measures; justify the selected methods and communicate results, assumptions, limitations, and evidence-based conclusions. The 6 points shown here assess the brief exam response; the report provides the broader C10 evidence.